



9 March 2026

CERTIFICATE OF ANALYSIS

Customer: Northrop Grumman Systems Corporation

Purchase Order No: 5000679536 Part Number: MSRR4040-VAFR

Material: F3C 433/M65: 38%; 285AW; 50"

Specification: MSRR4040, Issue 4, dated November 2012.

Certificate of Conformance: Hexcel lot 92M0111527 has been manufactured and tested in conformance with specification MSRR4040, Issue 4, dated November 2012.

Lot No: 92M0111527 Date of Manufacture: 20 February 2026

Storage Life: 12 months from date of manufacture when stored at 0°.

Resin Lot No: 92M0111313 Manufactured by: Hexcel

Glass Lot No: 31W0027641 Manufactured by: Hexcel

- I. Prepreg Physical, Chemical and Mechanical Properties
See Certificate of Analysis Report.
- II. Individual Spool Physical Properties
See In Process Control Acceptance Testing Data.

Note: See Packing List for date of shipment, actual quantity and spools shipped.

Brandt Bond

Certificate of Analysis Report

Specification: **MSRR4040 Rev. Issue 4**
Configuration Number: **CN1301**
Material Type: **F3C 433/M65;38%;285AW;50"**
Lot Number: **92M0111527**
Order Number: **92M0111527**

Report Printed: 25-Mar-2026 9:27:22AM
Status: **PASS**
Approval: **TUCKERRR**

1. Tensile, SBS, and TG Panel

ASTM D8336 Nottingham Tack

Roll/Panel	Test/Result:	Requirements:	Values:
92M0111527001/001	Tack Force (lbf)		4.338 - 3.024 - 4.361
	Tack Force Avg (lbf)		3.908

Dimensional Fiber Volume

Roll/Panel	Test/Result:	Requirements:	Values:
92M0111527001/001	Laminate Density By Immersion Avg (g/cc)		1.5156
	Laminate Density By Immersion Std (g/cc)		0.0005
	Laminate Density By Immersion CoV		0.03
	Dimensional Fiber Volume (%)		54.39 - 54.68 - 54.57
	Dimensional Fiber Volume Avg (%)	50 - 56	55
	Dimensional Fiber Volume Std (%)		0.146
	Dimensional Fiber Volume CoV		0.27
	Void Volume (%)		1.0 - 1.1 - 1.1
	Void Volume Avg (%)	MAX= 2	1
	Void Volume Std (%)		0.045
	Void Volume CoV		4.27
92M0111527019/002	Laminate Density By Immersion Avg (g/cc)		1.5126
	Laminate Density By Immersion Std (g/cc)		0.0010
	Laminate Density By Immersion CoV		0.07
	Dimensional Fiber Volume (%)		54.56 - 54.41 - 54.12
	Dimensional Fiber Volume Avg (%)	50 - 56	54
	Dimensional Fiber Volume Std (%)		0.224
	Dimensional Fiber Volume CoV		0.41
	Void Volume (%)		1.3 - 1.2 - 1.2
	Void Volume Avg (%)	MAX= 2	1
	Void Volume Std (%)		0.031
	Void Volume CoV		2.49

Short Beam Shear Stress-Warp-Ambient

Roll/Panel	Test/Result:	Requirements:	Values:
92M0111527001/001	SBS Strength (MPa)		88.3 - 87.6 - 88.9 - 85.1 - 88.1

Certificate of Analysis Report

Specification: **MSRR4040 Rev. Issue 4**
 Configuration Number: **CN1301**
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Short Beam Shear Stress-Warp-Ambient (Continued)

Roll/Panel	Test/Result:	Requirements:	Values:
92M0111527001/001	SBS Strength Avg (MPa)	MIN= 65	88
	SBS Strength Std (MPa)		1.47
	SBS Strength CoV		1.68
92M0111527019/002	SBS Strength (MPa)		91.2 - 85.6 - 82.1 - 87.2 - 86.8
	SBS Strength Avg (MPa)	MIN= 65	87
	SBS Strength Std (MPa)		3.26
	SBS Strength CoV		3.76

Short Beam Shear Stress-Warp-150C(302F)

Roll/Panel	Test/Result:	Requirements:	Values:
92M0111527001/001	SBS Strength (MPa)		67.7 - 68.9 - 66.5 - 69.0 - 69.1
	SBS Strength Avg (MPa)	MIN= 50	68
	SBS Strength Std (MPa)		1.11
	SBS Strength CoV		1.63
92M0111527019/002	SBS Strength (MPa)		61.6 - 63.4 - 63.8 - 62.9 - 63.4
	SBS Strength Avg (MPa)	MIN= 50	63
	SBS Strength Std (MPa)		0.86
	SBS Strength CoV		1.36

Tg By DMA

Roll/Panel	Test/Result:	Requirements:	Values:
92M0111527001/001	Tg by DMA in °C (° C)		268.2
	Tg by DMA in °C Avg (° C)	MIN= 225.0	268.2
92M0111527019/002	Tg by DMA in °C (° C)		265.0
	Tg by DMA in °C Avg (° C)	MIN= 225.0	265.0

Calculated Thickness/Ply

Roll/Panel	Test/Result:	Requirements:	Values:
92M0111527001/001	Calculated Thickness/Ply Avg (mm)	0.28 - 0.32	0.30
	Calculated Thickness/Ply StDev (mm)		0.0025
	Calculated Thickness/Ply CoV		0.99
92M0111527019/002	Calculated Thickness/Ply Avg (mm)	0.28 - 0.32	0.30

Certificate of Analysis Report

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Calculated Thickness/Ply (Continued)

Roll/Panel	Test/Result:	Requirements:	Values:
92M0111527019/002	Calculated Thickness/Ply StDev (mm)		0.0025
	Calculated Thickness/Ply CoV		0.91

Tensile Stress, Modulus, and Elongation-Warp-Ambient

Roll/Panel	Test/Result:	Requirements:	Values:
92M0111527001/001	Norm Tensile Strength (MPa)		928.3 - 953.8 - 956.2 - 928.8 - 948.3
	Norm Tensile Strength Avg (MPa)	MIN= 700	943
	Norm Tensile Strength Std (MPa)		13.57
	Norm Tensile Strength CoV		1.44
	Norm Modulus (GPa)		117.4 - 120.1 - 126.1 - 122.2 - 126.2
	Norm Modulus Avg (GPa)	MIN= 55	122
	Norm Modulus Std (GPa)		3.820
92M0111527019/002	Norm Modulus CoV		3.12
	Norm Tensile Strength (MPa)		891.6 - 950.2 - 831.5 - 918.7 - 964.3
	Norm Tensile Strength Avg (MPa)	MIN= 700	911
	Norm Tensile Strength Std (MPa)		52.76
	Norm Tensile Strength CoV		5.79
	Norm Modulus (GPa)		130.0 - 120.4 - 126.9 - 125.0 - 124.9
	Norm Modulus Avg (GPa)	MIN= 55	125
	Norm Modulus Std (GPa)		3.496
	Norm Modulus CoV		2.78

2. PHYSICAL PROPERTIES

Volatile content @100°C in pan

Roll	Test/Result:	Requirements:	Values:
92M0111527001	Volatile content @100°C in a Pan (%)		0.5 - 0.6
	Volatile content @100°C in a Pan Avg (%)	MAX= 1	1
92M0111527019	Volatile content @100°C in a Pan (%)		0.5 - 0.5
	Volatile content @100°C in a Pan Avg (%)	MAX= 1	0

3. PHYSICAL TACK

Certificate of Analysis Report

Specification: **MSRR4040 Rev. Issue 4**
 Configuration Number: **CN1301**
 Material Type: **F3C 433/M65;38%;285AW;50"**
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Report Printed: 25-Mar-2026 9:27:22AM
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Tack

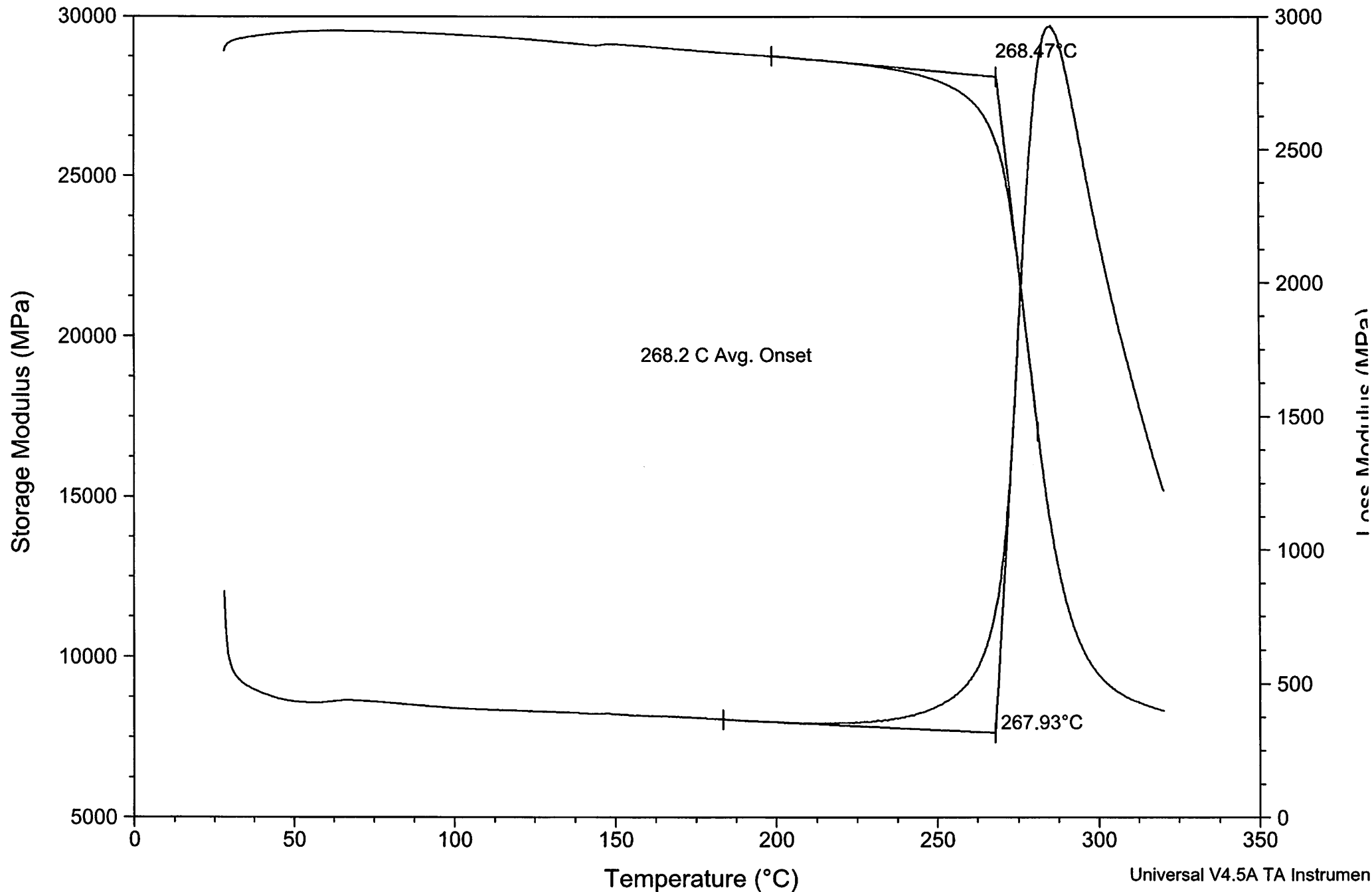
Roll	Test/Result:	Requirements:	Values:
92M0111527001	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111527003	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111527005	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111527007	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111527009	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111527011	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111527013	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111527015	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111527017	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0111527019	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30

End of report

Sample: 92M0111527-001 SP#001
Size: 34.9200 x 9.9300 x 2.3800 mm
Method: Temperature Ramp

DMA

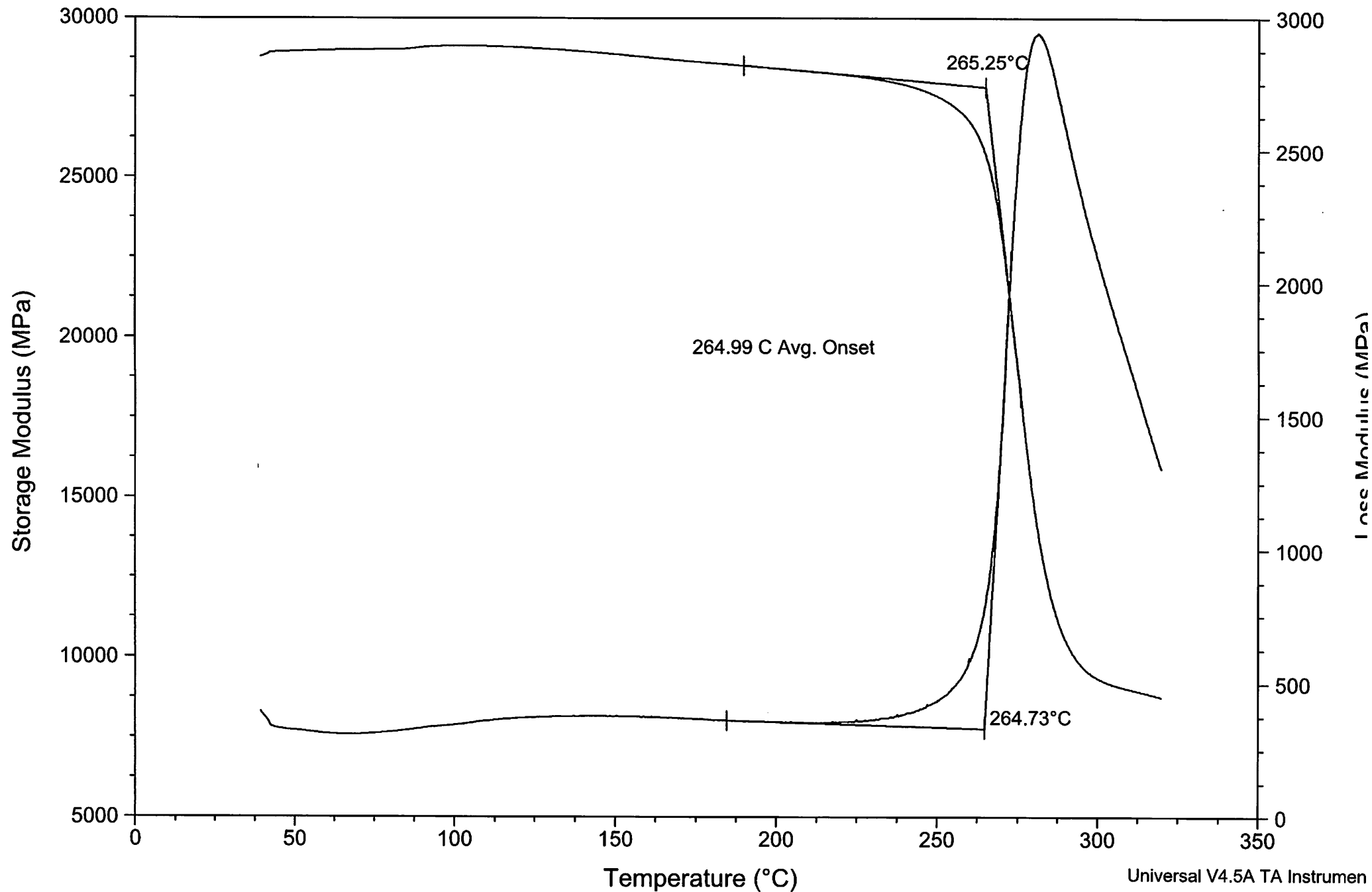
File: \\...\DMA Q800-2357\M65\92M0111527.001
Operator: KV
Run Date: 05-Mar-2026 14:40
Instrument: DMA Q800 V21.3 Build 96



Sample: 92M0111527-002 SP#019
Size: 34.9200 x 9.9300 x 2.4200 mm
Method: Temperature Ramp

DMA

File: \\...\\DMA Q800-2357\\M65\\92M0111527.002
Operator: KV
Run Date: 05-Mar-2026 15:54
Instrument: DMA Q800 V21.3 Build 96



IN PROCESS CONTROL

ACCEPTANCE TESTING DATA

RC / AWT Report for lot: **92M0111527**

3/6/2026

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Spool	Re - test	Rep	Init - Weight	Final Weight	RC	AWT	PAW	Width	Stat.	Tech	Date
92M 0111527001	0	1	4.6153	2.8593	38.0	286	461.5			HANSENAX	02/20/2026
		2	4.6226	2.8576	38.2	286	462.3			HANSENAX	02/20/2026
		3	4.6344	2.8641	38.2	286	463.4			HANSENAX	02/20/2026
		Avg.	4.6241		38	286	462	50.30	A	HANSENAX	02/20/2026
92M 0111527002	0	1	4.4817	2.8597	36.2	286	448.2			HANSENAX	02/20/2026
		2	4.4364	2.8345	36.1	283	443.6			HANSENAX	02/20/2026
		3	4.5291	2.8713	36.6	287	452.9			HANSENAX	02/20/2026
		Avg.	4.4824		36	286	448	50.25	A	HANSENAX	02/20/2026
92M 0111527003	0	1	4.5833	2.8367	38.1	284	458.3			HANSENAX	02/20/2026
		2	4.5995	2.8346	38.4	284	460.0			HANSENAX	02/20/2026
		3	4.5662	2.8246	38.1	282	456.6			HANSENAX	02/20/2026
		Avg.	4.5830		38	283	458	50.30	A	HANSENAX	02/20/2026
92M 0111527004	0	1	4.5999	2.8684	37.6	287	460.0			HANSENAX	02/20/2026
		2	4.6111	2.8807	37.5	288	461.1			HANSENAX	02/20/2026
		3	4.5905	2.8613	37.7	286	459.0			HANSENAX	02/20/2026
		Avg.	4.6005		38	287	460	50.30	A	HANSENAX	02/20/2026
92M 0111527005	0	1	4.5191	2.7799	38.5	278	451.9			HANSENAX	02/20/2026
		2	4.5354	2.7754	38.8	278	453.5			HANSENAX	02/20/2026
		3	4.5580	2.8149	38.2	282	455.8			HANSENAX	02/20/2026
		Avg.	4.5375		38	279	454	50.27	A	HANSENAX	02/20/2026
92M 0111527006	0	1	4.5276	2.8078	38.0	281	452.8			HANSENAX	02/20/2026
		2	4.5713	2.8109	38.5	281	457.1			HANSENAX	02/20/2026
		3	4.6858	2.8947	38.2	290	468.6			HANSENAX	02/20/2026
		Avg.	4.5949		38	284	460	50.25	A	HANSENAX	02/20/2026
92M 0111527007	0	1	4.5240	2.7959	38.2	280	452.4			HANSENAX	02/20/2026
		2	4.6545	2.8732	38.3	287	465.4			HANSENAX	02/20/2026
		3	4.5243	2.7960	38.2	280	452.4			HANSENAX	02/20/2026
		Avg.	4.5676		38	282	457	50.27	A	HANSENAX	02/20/2026
92M 0111527008	0	1	4.5544	2.8065	38.4	281	455.4			HANSENAX	02/20/2026
		2	4.6122	2.8396	38.4	284	461.2			HANSENAX	02/20/2026
		3	4.6205	2.8507	38.3	285	462.0			HANSENAX	02/20/2026
		Avg.	4.5957		38	283	460	50.30	A	HANSENAX	02/20/2026
92M 0111527009	0	1	4.5073	2.7919	38.1	279	450.7			HANSENAX	02/20/2026
		2	4.6098	2.8139	39.0	281	461.0			HANSENAX	02/20/2026
		3	4.5820	2.8094	38.7	281	458.2			HANSENAX	02/20/2026
		Avg.	4.5664		39	280	457	50.25	A	HANSENAX	02/20/2026
92M 0111527010	0	1	4.5130	2.7921	38.1	279	451.3			HANSENAX	02/20/2026
		2	4.4932	2.8511	36.5	285	449.3			HANSENAX	02/20/2026
		3	4.4862	2.7669	38.3	277	448.6			HANSENAX	02/20/2026
		Avg.	4.4975		38	280	450	50.25	A	HANSENAX	02/20/2026
92M 0111527011	0	1	4.5121	2.7942	38.1	279	451.2			HANSENAX	02/20/2026
		2	4.5909	2.8223	38.5	282	459.1			HANSENAX	02/20/2026
		3	4.5484	2.8291	37.8	283	454.8			HANSENAX	02/20/2026

IN PROCESS CONTROL

ACCEPTANCE TESTING DATA

RC / AWT Report for lot: **92M0111527**

3/6/2026

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Spool	Re - test	Rep	Init - Weight	Final Weight	RC	AWT	PAW	Width	Stat.	Tech	Date
=====											
			Avg. 4.5505		38	282	455	50.25	A	HANSENAX	02/20/2026
=====											
92M	0	1	4.5605	2.7967	38.7	280	456.0			HANSENAX	02/20/2026
0111527012											
		2	4.6484	2.8510	38.7	285	464.8			HANSENAX	02/20/2026
		3	4.6377	2.8739	38.0	287	463.8			HANSENAX	02/20/2026
			Avg. 4.6155		38	284	462	50.25	A	HANSENAX	02/20/2026
=====											
92M	0	1	4.4915	2.7618	38.5	276	449.2			HANSENAX	02/20/2026
0111527013											
		2	4.5979	2.8151	38.8	282	459.8			HANSENAX	02/20/2026
		3	4.5982	2.8220	38.6	282	459.8			HANSENAX	02/20/2026
			Avg. 4.5625		39	280	456	50.25	A	HANSENAX	02/20/2026
=====											
92M	0	1	4.4921	2.7805	38.1	278	449.2			CAZAREYX	02/20/2026
0111527014											
		2	4.5549	2.8215	38.1	282	455.5			CAZAREYX	02/20/2026
		3	4.5370	2.7984	38.3	280	453.7			CAZAREYX	02/20/2026
			Avg. 4.5280		38	280	453	50.25	A	CAZAREYX	02/20/2026
=====											
92M	0	1	4.5951	2.7903	39.3	279	459.5			CAZAREYX	02/20/2026
0111527015											
		2	4.5610	2.7754	39.1	278	456.1			CAZAREYX	02/20/2026
		3	4.5212	2.7638	38.9	276	452.1			CAZAREYX	02/20/2026
			Avg. 4.5591		39	278	456	50.25	A	CAZAREYX	02/20/2026
=====											
92M	0	1	4.5313	2.7906	38.4	279	453.1			CAZAREYX	02/20/2026
0111527016											
		2	4.5092	2.7959	38.0	280	450.9			CAZAREYX	02/20/2026
		3	4.4977	2.7659	38.5	277	449.8			CAZAREYX	02/20/2026
			Avg. 4.5127		38	278	451	50.26	A	CAZAREYX	02/20/2026
=====											
92M	0	1	4.5874	2.8290	38.3	283	458.7			CAZAREYX	02/20/2026
0111527017											
		2	4.6311	2.8459	38.5	285	463.1			CAZAREYX	02/20/2026
		3	4.5796	2.8132	38.6	281	458.0			CAZAREYX	02/20/2026
			Avg. 4.5994		38	283	460	50.25	A	CAZAREYX	02/20/2026
=====											
92M	0	1	4.5825	2.8345	38.1	283	458.2			CAZAREYX	02/20/2026
0111527018											
		2	4.6113	2.8957	37.2	290	461.1			CAZAREYX	02/20/2026
		3	4.5508	2.8550	37.3	286	455.1			CAZAREYX	02/20/2026
			Avg. 4.5815		38	286	458	50.60	A	CAZAREYX	02/20/2026
=====											
92M	0	1	4.6111	2.8711	37.7	287	461.1			CAZAREYX	02/20/2026
0111527019											
		2	4.6223	2.9038	37.2	290	462.2			CAZAREYX	02/20/2026
		3	4.6239	2.8898	37.5	289	462.4			CAZAREYX	02/20/2026
			Avg. 4.6191		38	289	462	50.32	A	CAZAREYX	02/20/2026
=====											
92M	0	1	4.5578	2.7932	38.7	279	455.8			CAZAREYX	02/20/2026
0111527020											
		2	4.6230	2.8396	38.6	284	462.3			CAZAREYX	02/20/2026
		3	4.5874	2.8010	38.9	280	458.7			CAZAREYX	02/20/2026
			Avg. 4.5894		39	281	459	50.30	A	CAZAREYX	02/20/2026
=====											
92M	0	1	4.6003	2.8613	37.8	286	460.0			CAZAREYX	02/20/2026
0111527021											
		2	4.6253	2.8633	38.1	286	462.5			CAZAREYX	02/20/2026
		3	4.5099	2.8106	37.7	281	451.0			CAZAREYX	02/20/2026
			Avg. 4.5785		38	284	458	50.25	A	CAZAREYX	02/20/2026
=====											
92M	0	1	4.5541	2.8114	38.3	281	455.4			CAZAREYX	02/20/2026
0111527022											
		2	4.6282	2.8504	38.4	285	462.8			CAZAREYX	02/20/2026

IN PROCESS CONTROL

ACCEPTANCE TESTING DATA

RC / AWT Report for lot: **92M0111527**

3/6/2026

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Spool	Re - test	Rep	Init - Weight	Final Weight	RC	AWT	PAW	Width	Stat.	Tech	Date
92M	0	3	4.5749	2.8263	38.2	283	457.5				
0111527022											
			Avg. 4.5857		38	283	459	50.30	A	CAZAREYX	02/20/2026
92M	0	1	4.5813	2.8273	38.3	283	458.1				
0111527023											
		2	4.5622	2.7996	38.6	280	456.2			CAZAREYX	03/02/2026
		3	4.5488	2.8121	38.2	281	454.9			CAZAREYX	03/02/2026
			Avg. 4.5641		38	281	456	50.30	A	CAZAREYX	03/02/2026

IN PROCESS CONTROL

ACCEPTANCE TESTING DATA

Lot Averages Report for Lot: **92M0111527**

03/06/2026

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Spool	Retest:	RC	AWT	PAW	Stamp
92M0111527001	0	38	286	462	A
92M0111527002	0	36	286	448	A
92M0111527003	0	38	283	458	A
92M0111527004	0	38	287	460	A
92M0111527005	0	38	279	454	A
92M0111527006	0	38	284	460	A
92M0111527007	0	38	282	457	A
92M0111527008	0	38	283	460	A
92M0111527009	0	39	280	457	A
92M0111527010	0	38	280	450	A
92M0111527011	0	38	282	455	A
92M0111527012	0	38	284	462	A
92M0111527013	0	39	280	456	A
92M0111527014	0	38	280	453	A
92M0111527015	0	39	278	456	A
92M0111527016	0	38	278	451	A
92M0111527017	0	38	283	460	A
92M0111527018	0	38	286	458	A
92M0111527019	0	38	289	462	A
92M0111527020	0	39	281	459	A
92M0111527021	0	38	284	458	A
92M0111527022	0	38	283	459	A
92M0111527023	0	38	281	456	A
Averages:		38.1	282.6	456.0	



Product Certification

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COA Number:	SEG-COA-29673	Prod Order:	DOM:	Description:
Version:	1	31W0027641	27-Sep-2025	F3C433(FBL2)-50"
Issue Date:	09-Feb-2026	Weave Pattern:	5 Harness Satin	

Specifications

F3C433(FBL2) - GREIGE - 50inch - HRMS 111546 Rev. 6 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

F3C433(FBL2) - GREIGE - 50inch - MSRR4040 Issue 4 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

F3C433(FBL2) - GREIGE - 50inch - MSRR9992 Issue 12 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

Rolls Tested

Fabric Testing Results

Serial#:	Prod Order:	Property:	Specification:	Result:
31G0000367014	31W0027641	Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	285
31G0000367015		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000367016		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Dried FAW Avg	278 - 292 g/m ²	285
31G0000367017		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000367018		Width	49.5 - 50.5 in	50.1
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7

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Michael Galvan
Certification Technician



Product Certification

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31G0000367018	31W0027641	Dried FAW Avg	278 - 292 g/m ²	283
31G0000367019		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000367020		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367021		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	285
31G0000367022		Width	49.5 - 50.5 in	50.1
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000367023		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367024		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000367025		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367026		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.8

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31G0000367026	31W0027641	Dried FAW Avg	278 - 292 g/m ²	284
31G0000367027		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000367028		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367029		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.9
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367030		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Dried FAW Avg	278 - 292 g/m ²	285
31G0000367031		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.9
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367059		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	281
31G0000367060		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000367061		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.9
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7

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31G0000367061	31W0027641	Dried FAW Avg	278 - 292 g/m ²	282
31G0000367062		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000367063		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000367064		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000367065		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	282

Certification Exceptions/Additional Information

QMS07045, REV. 1, SEC. 5E

WARP: 250120210
FILL: 250120210

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Product Certification

Page 1 of 3

COA Number:	SEG-COA-29673	Prod Order:	DOM:	Description:
Version:	1	31W0027641	27-Sep-2025	F3C433(FBL2)-50"
Issue Date:	09-Feb-2026	Weave Pattern:	5 Harness Satin	

Specifications

F3C433(FBL2) - GREIGE - 50inch - HRMS 111546 Rev. 6 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

F3C433(FBL2) - GREIGE - 50inch - MSRR4040 Issue 4 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

F3C433(FBL2) - GREIGE - 50inch - MSRR9992 Issue 12 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

Rolls Tested

Fabric Testing Results

Serial#:	Prod Order:	Property:	Specification:	Result:
31G0000367041	31W0027641	Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367042		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367043		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	286
31G0000367044		Width	49.5 - 50.5 in	50.1
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367045		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7

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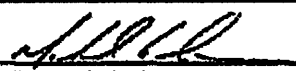
Product Certification

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31G0000367045	31W0027641	Dried FAW Avg	278 - 292 g/m ²	283
31G0000367046		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000367047		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367048		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000367049		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000367050		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367051		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000367052		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.9
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367053		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7

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31G0000367053	31W0027641	Dried FAW Avg	278 - 292 g/m ²	283
31G0000367054		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.6
		Dried FAW Avg	278 - 292 g/m ²	284
31G0000367055		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000367056		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000367066		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000367067		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Dried FAW Avg	278 - 292 g/m ²	281

Certification Exceptions/Additional Information

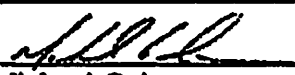
QMS07045, REV. 1, SEC. 5E

WARP: 250120210

FILL: 250120210

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**TEIJIN CARBON AMERICA, INC.**

121 Cardiff Valley Road
Rockwood, TN 37854
Tel. (865) 354-8408
Fax (865) 354-8409

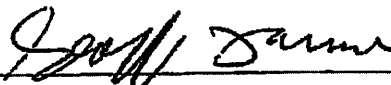
Certificate of Analysis

QAP 004 F2 Rev M

Customer: Hexcel Reinforcement Corp (TX)	Contact: Patrick Spieckermann; Amy Shehane; Jerry Tschoepe; Greg Early	Email/Fax: Patrick.Spieckermann@hexcel.com; amy.shehane@hexcel.com; jerry.tschoepe@; Greg.Early@
CustomerPO#: 31PO0014263	Order: SH001734	Ship Date: 7/1/2025
Customer P/N: n/a	Product: Tenax™-E HTA40 E13 3K 200tex	
Part Number: G305303	Quantity: 10000.066 lb	Certifies To: MS-X-CTC-105 Rev B; MSRR9317/11 Issue 1
Lot Number: 250120210		

We are pleased to advise that the product (s) listed above has been released for shipment. We certify that this lot of Carbon Fiber has been tested by standard test methods and determined to have the following properties:

	<u>Average</u>	<u>CV%</u>
Tensile Strength, MPA (ksi):	4145.8 (601)	4.2
Tensile Modulus, GPa (msi):	238.4 (34.6)	0.8
Elongation, %:	1.64	4.0
Linear Density (without size), g/m:	0.1994	1.0
Size, %:	1.27	7.3
Fiber Density, g/cc:	1.77	
Filament Diameter, µm:	7	
Twist, turn/m:	Never Twisted	
Date of Manufacture:	January 2025	
Manufacturing Location:	Oberbruch, Heinsberg, Germany	


Quality Assurance 6/30/2025

These commodities, technology or software are controlled by U. S. Export Administration Regulation. Diversion contrary to U. S. Law is prohibited.

If goods are not received in a reasonable time or if there are any questions, please call 1-800-252-3001 (or (865) 354-8408).
Teijin Carbon America, Inc. 121 Cardiff Valley Road, Rockwood, TN 37854.

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527001	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
47	50	2	0.7	RESIN LEAN

Final Footage	Net Feet	Credit Method	Defect %	Date
58	57	A	0.287%	3/2/2026

Inspection Stamp	Date
	3/2/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527002	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
7	50	0	0.0	FUZZBALL
12	50	24	8.3	RESIN LEAN
20	50	0	0.0	FUZZBALL
25	50	24	8.3	RESIN LEAN
53	50	0	0.0	FUZZBALL
84	50	0	0.0	FUZZBALL
85	50	0	0.0	FUZZBALL
93	50	0	0.0	FUZZBALL
98	50	0	0.0	FUZZBALL
103	50	0	0.0	FUZZBALL
109	50	0	0.0	FUZZBALL
113	50	0	0.0	FUZZBALL
116	50	0	0.0	FUZZBALL
119	50	0	0.0	FUZZBALL
168	50	36	12.5	RESIN LEAN

Final Footage	Net Feet	Credit Method	Defect %	Date
175	168	A	4.000%	2/22/2026

Inspection Stamp	Date
	2/22/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527003	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
30	50	24	8.3	CREASE
58	50	0	0.0	FUZZBALL
81	50	0	0.0	FUZZBALL
82	50	0	0.0	FUZZBALL
88	50	0	0.0	FUZZBALL
93	50	0	0.0	FUZZBALL
147	50	0	0.0	FUZZBALL
159	50	0	0.0	FUZZBALL
162	50	24	8.3	RESIN LEAN
190	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
193	189	A	2.073%	3/2/2026

Inspection Stamp	Date
	3/2/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527005	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
20	50	24	8.3	RESIN LEAN
21	50	0	0.0	FUZZBALL
52	50	84	29.2	WRINKLES
84	50	0	0.0	FUZZBALL
99	50	0	0.0	FUZZBALL
106	50	2	0.7	FUZZBALL
138	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
162	152	A	5.658%	3/2/2026

Inspection Stamp	Date
	3/2/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527006	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
21	50	2	0.7	SNAG
38	50	0	0.0	FUZZBALL
91	50	0	0.0	FUZZBALL
134	50	2	0.7	RESIN LEAN

Final Footage	Net Feet	Credit Method	Defect %	Date
200	199	A	0.167%	3/2/2026

Inspection Stamp	Date
	3/2/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527007	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
4	50	0	0.0	FUZZBALL
21	50	0	0.0	FUZZBALL
57	50	0	0.0	FUZZBALL
59	50	0	0.0	FUZZBALL
105	50	2	0.7	SNAG
121	50	0	0.0	FUZZBALL
135	50	0	0.0	FUZZBALL
167	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
219	218	A	0.076%	3/2/2026

Inspection Stamp	Date
	3/2/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527008	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
15	50	0	0.0	FUZZBALL
25	50	48	16.7	RESIN LEAN
31	50	24	8.3	RESIN LEAN
52	50	0	0.0	FUZZBALL
63	50	0	0.0	FUZZBALL
80	50	32	11.1	WRINKLES
88	50	0	0.0	FUZZBALL
101	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
113	104	A	7.670%	3/2/2026

Inspection Stamp	Date
	3/2/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527010	F3C433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
106	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
214	214	A	0.000%	2/20/2026

Inspection Stamp	Date
	2/20/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527011	F3C433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
105	50	4	1.4	RESIN LEAN

Final Footage	Net Feet	Credit Method	Defect %	Date
223	222	A	0.149%	2/20/2026

Inspection Stamp	Date
	2/20/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527012	F3C433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
56	50	1	0.3	RESIN LEAN
63	50	1	0.3	SLUB
72	50	0	0.0	FUZZBALL
134	50	2	0.7	FLOAT
144	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
218	217	A	0.153%	2/20/2026

Inspection Stamp	Date
	2/20/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527013	F3C433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
27	50	0	0.0	FUZZBALL
62	50	24	8.3	WRINKLES
87	50	36	12.5	WRINKLES
196	50	60	20.8	WRINKLES

Final Footage	Net Feet	Credit Method	Defect %	Date
225	215	A	4.444%	2/20/2026

Inspection Stamp	Date
	2/20/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527014	F3C433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
65	50	156	54.2	CURED RESIN

Final Footage	Net Feet	Credit Method	Defect %	Date
215	202	A	6.047%	2/20/2026

Inspection Stamp	Date
	2/20/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527015	F3C433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
87	50	2	0.7	FLOAT

Final Footage	Net Feet	Credit Method	Defect %	Date
223	222	A	0.075%	2/20/2026

Inspection Stamp	Date
	2/20/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527016	F3C433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
54	50	0	0.0	FUZZBALL
66	50	0	0.0	FUZZBALL
118	50	24	8.3	CURED RESIN

Final Footage	Net Feet	Credit Method	Defect %	Date
224	222	A	0.893%	2/20/2026

Inspection Stamp	Date
	2/20/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527017	F3C433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
114	50	0	0.0	FUZZBALL
148	50	1	0.3	SLUB
157	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
223	222	A	0.037%	2/21/2026

Inspection Stamp	Date
	2/21/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527018	F3C433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks

Final Footage	Net Feet	Credit Method	Defect %	Date
194	194	A	0.000%	2/21/2026

Inspection Stamp	Date
	2/21/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527019	F3C433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
69	50	0	0.0	FUZZBALL
78	50	0	0.0	FUZZBALL
94	50	2	0.7	FUZZBALL
127	50	2	0.7	FLOAT
196	50	1	0.3	SLUB

Final Footage	Net Feet	Credit Method	Defect %	Date
265	264	A	0.157%	2/21/2026

Inspection Stamp	Date
	2/21/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527020	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks

Final Footage	Net Feet	Credit Method	Defect %	Date
29	29	A	0.000%	2/22/2026

Inspection Stamp	Date
	2/22/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527021	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
12	50	0	0.0	FUZZBALL
47	50	0	0.0	FUZZBALL
48	50	0	0.0	FUZZBALL
57	50	2	0.7	FUZZBALL
62	50	1	0.3	FUZZBALL
69	50	24	8.3	FUZZBALL
87	50	0	0.0	FUZZBALL
107	50	0	0.0	FUZZBALL
116	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
134	131	A	1.679%	2/22/2026

Inspection Stamp	Date
	2/22/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527022	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
14	50	0	0.0	FUZZBALL
42	50	0	0.0	FUZZBALL
94	50	0	0.0	FUZZBALL

Final Footage	Net Feet	Credit Method	Defect %	Date
97	97	A	0.000%	2/22/2026

Inspection Stamp	Date
	2/22/2026

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0111527	92M0111527023	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
14	50	0	0.0	FUZZBALL
16	50	0	0.0	FUZZBALL
24	50	2	0.7	RESIN LEAN
41	50	2	0.7	RESIN LEAN

Final Footage	Net Feet	Credit Method	Defect %	Date
53	52	A	0.629%	3/2/2026

Inspection Stamp	Date
	3/2/2026